

EE605 Endsem Solutions

Puspabeethi Samanta, Suketu Patni

May 5, 2026

1 Question 1

Addition and multiplication tables of $\mathbb{F}_4 = \{0, 1, \gamma, \gamma^2\}$:

+	0	1	γ	γ^2	*	0	1	γ	γ^2
0	0	1	γ	γ^2	0	0	0	0	0
1	1	0	γ^2	γ	1	0	1	γ	γ^2
γ	γ	γ^2	0	1	γ	0	γ	γ^2	1
γ^2	γ^2	γ	1	0	γ^2	0	γ^2	1	γ

1.a

A primitive element of field $\mathbb{F}_4$ generates all the elements of the multiplicative group $\mathbb{F}_4 \setminus \{0\}$ of order 3. From the multiplication table,

- $\gamma^3 = \gamma \cdot \gamma^2 = 1$. Thus $\langle \gamma \rangle = \{\gamma, \gamma^2, 1\}$.
- $(\gamma^2)^2 = \gamma$; $(\gamma^2)^3 = (\gamma^2)^2 \cdot \gamma^2 = \gamma \cdot \gamma^2 = 1$. Thus $\langle \gamma^2 \rangle = \{\gamma^2, \gamma, 1\}$.

Since γ, γ^2 (non-identity elements of multiplicative group) generates $\mathbb{F}_4 \setminus \{0\}$, these are the primitive elements.

1.b

The codewords of $RS_4(\{1, \gamma, \gamma^2\}, n=3, k=2)$ correspond to polynomials $f(x) = a_0 + a_1x$ (degree $< k$). A codeword of the code is $(f(1), f(\gamma), f(\gamma^2))$. Since the 1st coordinate is 1, therefore $f(1) = a_0 + a_1 = 1 \implies a_0 = 1 - a_1 = 1 + a_1$ ($-a = a$ in $\mathbb{F}_4$ arithmetic, for $a \in \mathbb{F}_4$). Therefore, $f(x) = (1 + a_1) + a_1x$.

Next, we obtain codewords $(f(1), f(\gamma), f(\gamma^2))$ for each value of $a_1 \in \mathbb{F}_4$.

- $a_1 = 0 : f(\gamma) = 1, f(\gamma^2) = 1 \implies \text{codeword} = (1, 1, 1)$.
- $a_1 = 1 : f(\gamma) = \gamma, f(\gamma^2) = \gamma^2 \implies \text{codeword} = (1, \gamma, \gamma^2)$.
- $a_1 = \gamma : f(\gamma) = 0, f(\gamma^2) = \gamma \implies \text{codeword} = (1, 0, \gamma)$.
- $a_1 = \gamma^2 : f(\gamma) = \gamma^2, f(\gamma^2) = 0 \implies \text{codeword} = (1, \gamma^2, 0)$.

1.c

A codeword of $RS_4(\{1, \gamma, \gamma^2\}, n=3, k=2)$ is $(f(1), f(\gamma), f(\gamma^2))$ where $f(x) = a_0 + a_1x$.

Consider all cases where $a_1 = 0$. Then $f(x) = a_0 \forall x \in \{1, \gamma, \gamma^2\}$. For such cases, $(0, 0, 0)$ and $(1, 1, 1)$ are the possible codewords in $\{0, 1\}^3$.

Next, consider the remaining cases where $a_1 \neq 0$. For a codeword to lie in $\{0, 1\}^3$, at least two of its symbols must be equal. i.e. $f(x_1) = f(x_2)$ where $x_1 \neq x_2$, and $a_1 \neq 0$. This gives $a_0 + a_1x_1 = a_0 + a_1x_2 \implies a_1x_1 = a_1x_2 \implies x_1 = x_2$, which is not possible since $x_1 \neq x_2$.

Therefore $(0, 0, 0)$ and $(1, 1, 1)$ are the only codewords of $RS_4(\{1, \gamma, \gamma^2\}, n=3, k=2)$ contained in $\{0, 1\}^3$.

2 Question 2

1. **Hamming:** It states that

$$A_q(n, d) \leq \frac{q^n}{V_q(n, t)}$$

where $A_q(n, d) = \max\{|C| : C \subseteq \mathbb{F}_q^n \text{ is a (not necessarily linear) code of length } n \text{ and } d(C) = d\}$, and

$$V_q(n, t) = \sum_{i=0}^t \binom{n}{i} (q-1)^i$$

where $t = \lfloor \frac{d-1}{2} \rfloor$. For our case, $n = 10, d = 5, q = 2$ and C happens to be linear. So, $|C| \leq A_2(10, 5) \leq \frac{2^{10}}{V_2(10, 2)}$. We have

$$V_2(10, 2) = \sum_{i=0}^2 \binom{10}{i} (2-1)^i = \binom{10}{0} + \binom{10}{1} + \binom{10}{2} = 1 + 10 + 45 = 56$$

So $2^k = |C| \leq \frac{1024}{56} = \frac{128}{7} \Rightarrow k \leq \log_2(\frac{128}{7}) \Rightarrow k \leq 4$ (since k is an integer).

2. **Gilbert-Varshamov:** It states that

$$A_q(n, d) \geq \frac{q^n}{V_q(n, d-1)} \Rightarrow 2^k \geq \frac{2^{10}}{V_2(10, 4)}$$

We have

$$V_2(10, 4) = \sum_{i=0}^4 \binom{10}{i} (2-1)^i = \binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3} + \binom{10}{4} = 1 + 10 + 45 + 120 + 210 = 56 + 330 = 386$$

So $2^k \geq 1024/386 \Rightarrow k \geq 2$. The best value of k is atleast 2.

3. **Plotkin:** We have, for $2d \leq n$,

$$A_2(n, d) \leq 2d \times 2^{n-2d+1} \Rightarrow 2^k \leq 2 \times 5 \times 2^{10-10+1} = 20 \Rightarrow k \leq 4$$

4. **Singleton:** $k \leq n - d + 1 = 6$

So, using all these bounds, we conclude that $k \in \{2, 3, 4\}$.

3 Question 3

C is an (n, k, d) linear code over $GF(q)$. We need to prove that C is an MDS code $\implies$ a minimum weight codeword lies in any d -sized subset of coordinates, and vice-versa.

- Let C be an MDS code with generator matrix G . Then $d = n - k + 1$. Let $S \subset [n]$, with $|S| = d$. Consider S^c of size $|S^c| = n - d = k - 1$. A codeword of C is $c = [u \cdot g_1 \quad u \cdot g_2 \quad \cdots \quad u \cdot g_n]$, where $u \in GF(q)^k$, and $G = [g_1 \quad g_2 \quad \cdots \quad g_n]$.

Let G_{S^c} be the $k \times (k-1)$ submatrix of G indexed by entries of S^c . We need to show that $\exists$ a non-zero codeword $c \in C$ s.t. $uG_{S^c} = 0$. For an MDS code, any k columns of the generator matrix $G = [g_1 \quad g_2 \quad \cdots \quad g_n]$ of C are linearly independent. By rank-nullity, $\dim(\text{nullspace}(G_{S^c})) = k - \text{rank}(G_{S^c}) = k - (k-1) = 1$. Therefore, $\exists u \in GF(q)^k$ with $c_i = u_i \cdot g_i = 0 \quad \forall i \in S^c \implies \text{support}(c) \subseteq S$. Since $|S| = d$, $\text{support}(c) = S$, and c is a minimum-weight codeword.

- Next, we assume that for every d -sized subset S of coordinates, $\exists$ a codeword c of weight d with $\text{support}(c) = S$. Then for every possible choice of d columns in the parity check matrix H , there is a linear combination that equals zero. So, H cannot have d linearly independent columns. Therefore, $\text{rank}(H) \leq d-1 \implies n-k \leq d-1 \implies d \geq n-k+1$. From Singleton bound, we have $d \leq n-k+1$. Therefore, $d = n-k+1$ and C is an MDS code.

4 Question 4

4.a

Clearly, $n = 6$ and $k = 3$. A parity check matrix for C is

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

The first two columns of H are the same, so they are linearly dependent. Further, H has no zero column. So $d(C) = 2$.

4.b

To detect upto two errors and correct one error simultaneously, $d \geq 2 + 1 + 1 = 4$. So let us try and make $d = 4$. Currently, $d = 2$. So clearly, we need to apply two operations on it. But we can't change k i.e. the number of rows of G . Instead we can add two columns to G (so the operation used is extension). so that d becomes 4. So say we add them as follows.

$$G' = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & b_1 & b_4 \\ 0 & 1 & 0 & 1 & 1 & 0 & b_2 & b_5 \\ 0 & 0 & 1 & 0 & 1 & 1 & b_3 & b_6 \end{pmatrix}$$

Instead of trying to compute d using the parity check matrix, we can enumerate all the codewords individually, since there are only 8 of them.

1. 00000000
2. 100110 b_1b_4
3. 010110 b_2b_5
4. 001011 b_3b_6
5. 110000($b_1 + b_2$)($b_4 + b_5$)
6. 011101($b_2 + b_3$)($b_5 + b_6$)
7. 101101($b_1 + b_3$)($b_4 + b_6$)
8. 111011($b_1 + b_2 + b_3$)($b_4 + b_5 + b_6$)

The inferences we can draw for each nonzero codeword having weight atleast 4 are as follows.

1. Atleast one of b_1, b_4 must be one.
2. Atleast one of b_2, b_5 must be one.
3. Atleast one of b_3, b_6 must be one.
4. $b_1 + b_2 = b_4 + b_5 = 1$

We may simply choose $b_1 = 1, b_2 = 0, b_3 = 1, b_4 = 0, b_5 = 1, b_6 = 1$. This gives us

$$G' = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

with $(n', k', d') = (8, 3, 4)$.

4.c

The solution in part (b) will work since it is already able to correct one error therein. It has been said that the system is willing to reduce the number of messages, not that doing that is required. A solution which involves shortening at the first coordinate also works, and in fact, this gives better rate. Indeed, shortening at the first (or second) coordinate removes the corresponding column from H and gives

$$H'' = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Here it is easily seen that no two columns are the same (and trivially the rank is 3), so $d'' = 3$. We also get $n'' = 5$ and $n'' - k'' = 3 \Rightarrow k'' = 2$. Since this is of the form $[-A^T \mid I_{n''-k'']}]$, a generator matrix for the code is

$$G'' = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

4.d

The solution in (b) would have worked here as well, except that G has too few rows. So we could simply have added a row to G . But augmentation is not allowed. So we lengthen instead (we add both another column and another row to G). We need to be able to detect atleast 1 error, so $d \geq 2$. An easy thing to do is to add the all zeroes column to G and then add another row so that the resulting code has no codeword of weight 1 (the current minimum distance is 2). As enumerated in part (b), we see that the first 5 coordinates of each nonzero codeword of C contain atleast 2 '1's. This implies that

$$G''' = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

will work since any combination of the last row with the other rows does not change the first 5 coordinates of the codeword, and just the last row by itself also has weight 2. Here, clearly, $(n''', k''', d''') = (7, 4, 2)$.

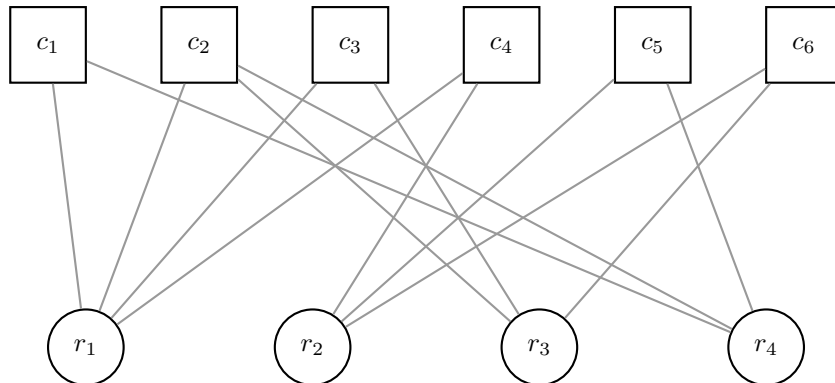
5 Question 5

5.a

Let the rows of

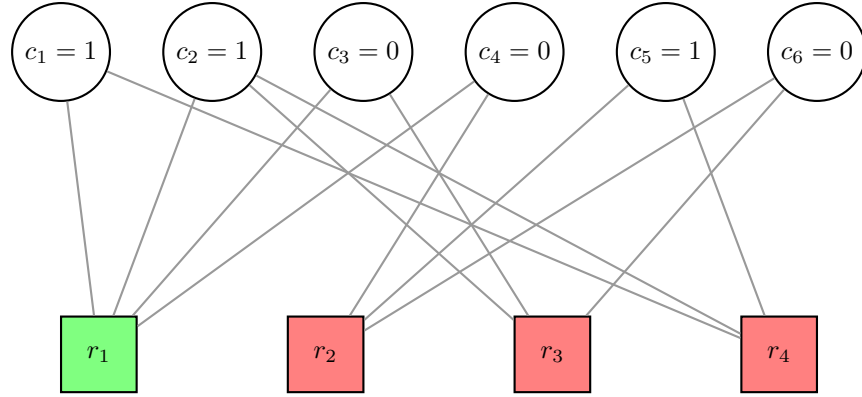
$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

be r_1, r_2, r_3, r_4 and its columns (or variables) be $c_1, c_2, c_3, c_4, c_5, c_6$. Then the Tanner graph for H is as below

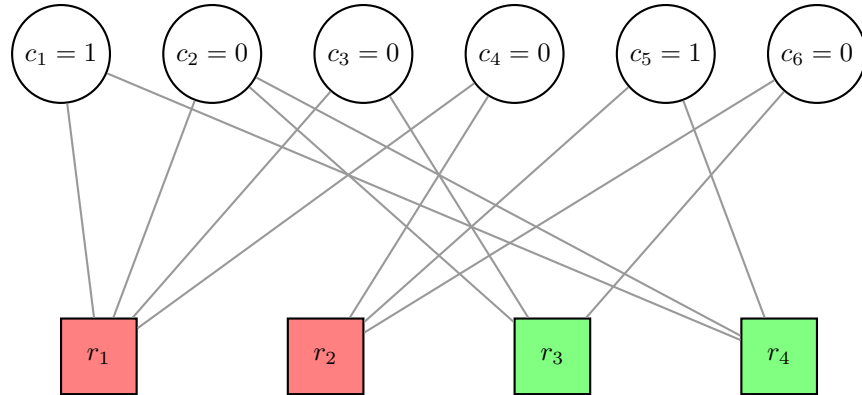


5.b

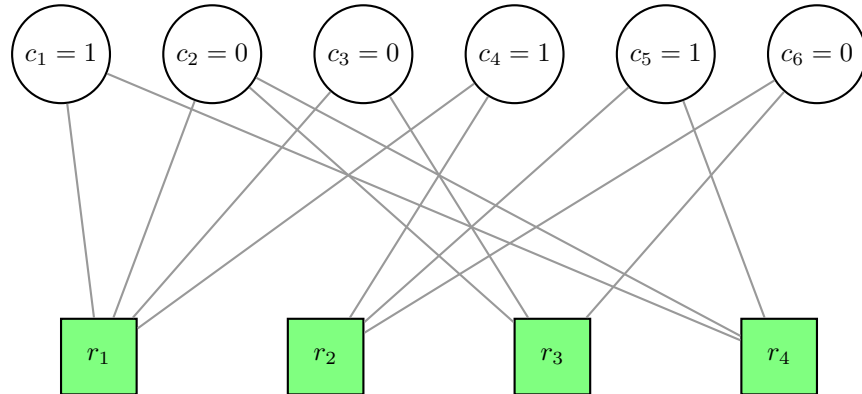
As given in the question, first we determine the set of rows for which the parity check given by that row fails for the received codeword. We have $r_1 \cdot y^{(0)} = 0$, $r_2 \cdot y^{(0)} = 1$, $r_3 \cdot y^{(0)} = 1$, and $r_4 \cdot y^{(0)} = 1$. So we get the following graph, where a red check node implies non-satisfaction.



Now c_1, c_3 , and c_4 have one unsatisfied neighbour, whereas the rest have 2. Further, since there is only one satisfied parity check, each column with 2 unsatisfied neighbours must have more unsatisfied neighbours than satisfied neighbours. Among c_2, c_5, c_6 , we flip c_2 as it has the lowest variable index. So the new codeword is $y^{(1)} = (1, 0, 0, 0, 1, 0)$. Now note that $r_1 \cdot y^{(1)} = 1$, $r_2 \cdot y^{(1)} = 1$, $r_3 \cdot y^{(1)} = 0$, and $r_4 \cdot y^{(1)} = 0$.



Now c_3 has one unsatisfied neighbour, and c_4 has two. So we flip c_4 . Now finally we get $r_1 \cdot y^{(2)} = 0$, $r_2 \cdot y^{(2)} = 0$, $r_3 \cdot y^{(2)} = 0$, and $r_4 \cdot y^{(2)} = 0$.



So the decoded codeword is $y^{(2)} = (1, 0, 0, 1, 1, 0)$

6 Question 6

The set of all valid codewords is easily found to be $C = \{(0, 0, 0, 0), (1, 0, 0, 1), (1, 1, 1, 0), (0, 1, 1, 1)\}$. Immediately we note that the minimum distance is 2, and so the code cannot correct any error. Indeed, the received codeword is equidistant from both $(0, 0, 0, 0)$ and $(1, 0, 0, 1)$. So we expect two answers for this. We find that our answer depends on our choices of coset leaders.

6.a

Coset Leader (e)	Codewords $+e$		
0000	1001	1110	0111
1000	0001	0110	1111
0100	1101	1010	0011
0010	1011	1100	0101

Alternatively,

Coset Leader (e)	Codewords $+e$		
0000	1001	1110	0111
0001	1000	1111	0110
0010	1011	1100	0101
0100	1101	1010	0011

6.b

Syndrome (S)	Coset Leader (e)
00	0000
10	1000
01	0100
11	0010

Alternatively,

Syndrome (S)	Coset Leader (e)
00	0000
10	0001
01	0100
11	0010

6.c

The syndrome of the received codeword is $s = Hy^T = (1, 0)^T$. So the error is either $(0, 0, 0, 1)$ or $(1, 0, 0, 0)$. So the decoded codeword $(y - e)$ is either $(0, 0, 0, 0)$ or $(1, 0, 0, 1)$.

7 Question 7

$C \subseteq \mathbb{F}_2^n$ is a binary linear code. 1-variable weight enumerator : $A_C(z) = \sum_{w=0}^n A_w z^w$, where $A_w = \sum_{c \in C} \mathbf{1}_{\{wt(c)=w\}}$.
 2-variable weight enumerator : $W_C(x, y) = \sum_{c \in C} x^{n-wt(c)} y^{wt(c)}$.

$$(MW1) \quad A_{C^\perp}(z) = \frac{1}{|C|} (1+z)^n A_C\left(\frac{1-z}{1+z}\right)$$

$$(MW2) \quad W_{C^\perp}(x, y) = \frac{1}{|C|} W_C(x+y, x-y)$$

7.a

$$\begin{aligned} W_C(x, y) &= \sum_{c \in C} x^{n-wt(c)} y^{wt(c)} \\ &= x^n \sum_{c \in C} \left(\frac{y}{x}\right)^{wt(c)} = x^n \sum_{w=0}^n A_w \left(\frac{y}{x}\right)^w = x^n A_C \left(\frac{y}{x}\right) \end{aligned}$$

7.b

$$(MW2) \quad W_{C^\perp}(x, y) = \frac{1}{|C|} W_C(x + y, x - y)$$

Substituting $x = 1, y = z$,

$$W_{C^\perp}(1, z) = \frac{1}{|C|} W_C(1 + z, 1 - z)$$

Using part 7.a, the following result is obtained.

$$\frac{1}{|C|} W_C(1 + z, 1 - z) = \frac{1}{|C|} (1 + z)^n A_C \left(\frac{1 - z}{1 + z}\right)$$

Since $A_{C^\perp}(z) = W_{C^\perp}(1, z)$, therefore

$$A_{C^\perp}(z) = \frac{1}{|C|} (1 + z)^n A_C \left(\frac{1 - z}{1 + z}\right) \quad (MW1)$$

7.c

$$(MW1) \quad A_{C^\perp}(z) = \frac{1}{|C|} (1 + z)^n A_C \left(\frac{1 - z}{1 + z}\right)$$

Using part 7.a, the following result is obtained.

$$\frac{1}{|C|} (1 + z)^n A_C \left(\frac{1 - z}{1 + z}\right) = \frac{1}{|C|} W_C(1 + z, 1 - z)$$

Substituting $z = \frac{y}{x}$, and then simplifying,

$$x^{-n} \frac{1}{|C|} (x + y)^n A_C \left(\frac{x - y}{x + y}\right) = x^{-n} \frac{1}{|C|} W_C(x + y, x - y)$$

Since $W_{C^\perp}(x, y) = x^n A_{C^\perp} \left(\frac{y}{x}\right)$ (obtained by applying part 7.a result to the dual code $C^\perp$), therefore

$$W_{C^\perp}(x, y) = \frac{1}{|C|} W_C(x + y, x - y) \quad (MW2)$$

7.d

Generator matrix of given binary linear code is $G = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$

7.d.1

Given $[n = 4, k = 2]$ code (say C) has following $2^k = 4$ codewords : $\{0000, 1100, 0111, 1011\}$ with $A_0 = 1, A_2 = 1$, and $A_3 = 2$.

$$\begin{aligned} W_C(x, y) &= \sum_{c \in C} x^{n-wt(c)} y^{wt(c)} = \sum_{w=0}^n A_w x^{n-w} y^w \\ &= x^4 + x^2 y^2 + 2x y^3 \end{aligned}$$

7.d.2

By applying $\text{Row}_1(G) \rightarrow \text{Row}_1(G) + \text{Row}_2(G)$, we obtain the following generator matrix in systematic form,

$$G_{sys} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} = [I_2|A]$$

Corresponding parity check matrix H (also generator matrix of dual code $C^\perp$) is

$$G^\perp = H = [-A|I_2] = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Therefore $C^\perp$ has the following $2^{n-k} = 4$ codewords : $\{0000, 1110, 1101, 0011\}$ with $A_0 = 1, A_2 = 1$, and $A_3 = 2$.

$$W_{C^\perp}(x, y) = \sum_{c \in C^\perp} x^{n-wt(c)} y^{wt(c)} = x^4 + x^2 y^2 + 2xy^3$$

7.d.3

$$(MW2) \quad W_{C^\perp}(x, y) = \frac{1}{|C|} W_C(x + y, x - y)$$

From part 7.d.2,

$$W_{C^\perp}(x, y) = x^4 + x^2 y^2 + 2xy^3$$

Next, $|C| = 4$, and

$$\begin{aligned} \frac{1}{|C|} W_C(x + y, x - y) &= \frac{1}{4} [(x + y)^4 + (x + y)^2(x - y)^2 + 2(x + y)(x - y)^3] \\ &= \frac{1}{4} [(x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4) + (x^4 - 2x^2y^2 + y^4) + 2(x + y)(x^3 - 3x^2y + 3xy^2 - y^3)] \\ &= \frac{1}{4} [(x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4) + (x^4 - 2x^2y^2 + y^4) + 2(x + y)(x^4 - 3x^2y + 3xy^2 - y^3)] \\ &= \frac{1}{4} \cdot 4(x^4 + x^2y^2 + 2xy^3) = W_{C^\perp}(x, y) \end{aligned}$$

8 Question 8

8.a

Let the original code C have parameters (n, k, d) . The new length is clearly $n' = n + 1$. The number of codewords now is twice the original number, so the new dimension is $k' = k + 1$. Lastly, the codeword with all zeroes in the first n coordinates and a 1 in the last coordinate is in C , so $d' = 1$.

8.b

Fix $y \in \mathbb{F}_2^{n+1}$, and consider the set $B(y, pn) \cap C'$. This is the set

$$\begin{aligned} \{c \in C' : d(c, y) \leq pn\} &= \{c \in C' : d((c_1, \dots, c_n), (y_1, \dots, y_n)) + d((c_{n+1}), (y_{n+1})) \leq pn\} \\ &\subseteq \{c \in C' : d((c_1, \dots, c_n), (y_1, \dots, y_n)) \leq pn\} \\ &= \{(c, b) : c \in B((y_1, \dots, y_n), pn) \cap C, b \in \mathbb{F}_2\} \end{aligned}$$

So we directly get $|B(y, pn) \cap C'| \leq 2 \times |B((y_1, \dots, y_n), pn) \cap C| \leq 2L$. We are done.

8.c

The new code has minimum distance equal to 1, which is the minimum possible minimum distance for a nontrivial code. Yet it has non-trivial list-decodability. So clearly a large minimum distance is not necessary for the same.